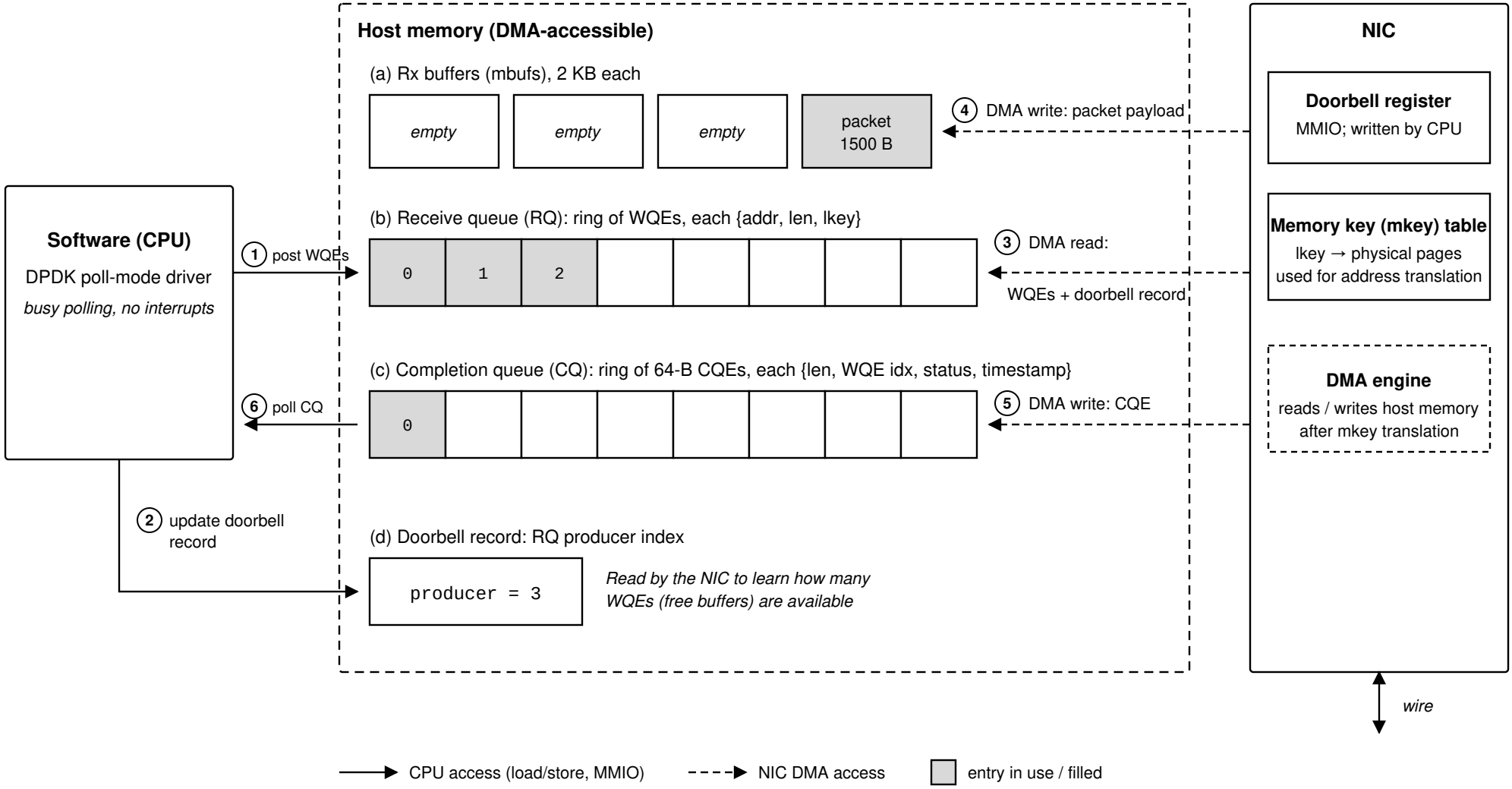
Host memory (DMA-accessible)
(a) Rx buffers (mbufs), 2 KB each
empty
empty
empty
packet
1500 B
4 DMA write: packet payload
NIC
Doorbell register
MMIO; written by CPU
Memory key (mkey) table
lkey → physical pages
used for address translation
DMA engine
reads / writes host memory
after mkey translation
Software (CPU)
DPDK poll-mode driver
busy polling, no interrupts
1 post WQEs
(b) Receive queue (RQ): ring of WQEs, each {addr, len, lkey}
0
1
2
3 DMA read:
WQEs + doorbell record
(c) Completion queue (CQ): ring of 64-B CQEs, each {len, WQE idx, status, timestamp}
6 poll CQ
0
5 DMA write: CQE
2 update doorbell record
(d) Doorbell record: RQ producer index
producer = 3
Read by the NIC to learn how many
WQEs (free buffers) are available
wire
CPU access (load/store, MMIO)
NIC DMA access
entry in use / filled